Progress Report Template

PROGRESS REPORTS ARE SUBMITTED INDIVIDUALLY BY EACH STUDENT IN THE GROUP

Save File as pdf with file name **FirstNameLastInitial_ProgressReport#.pdf** (e.g., JohnS_ProgressReport1.pdf if your name is John Smith)

Progress Report Number (1-4)

- **Work completed by each person must be checked into GitHub Repository under Implementation folder regularly. You may create additional folders/files as needed in the repository.**
- **Progress Report from each student must be checked into the GitHub repository under DocumentsAndReports.**
- **Progress Report from each student must be submitted on Blackboard**

The purpose of the Progress Report (along with implementation repo check in with work completed thus far) is to have an accountability for ongoing work through out the term. You need to create work logs with the following table:

Work Date/Hours Logs (all logs must be entered as and when work is completed) for EACH STUDENT. Having smaller chunks more regularly is important.

Creating broad logs with 10 hours and 12 hours in a day will be counted as inaccurate logging with insufficient granularity of work logs.

Instead create an entry for work done every day that you do the work.

Date	Number of Hours	Description of work done
October,3,2026	3	Made research with oke in order to find the site which we would get the logs and cves from inorder to use for ml training. Ended with discovering CIC-IDS-2017 and its logs which came in pcap and csvs. We saw that we would end up ignoring the pcap files as the csvs had all the info we need.

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Oct,7,2026	1.5	Worked on making the ml training repo along with help from ai

Summary Description of work done during this reporting period;

As of right now, weve found the source we will use as dummy data which is CIC-IDS-2017. It is a compilation of logs gooten from a company over 5 days with gbs of data within those 5 days and is more than enough for the training we will have to do. One of the initial road blocks we faced was getting the data, then how to host it inexpensively which we saw would be to host it in the cloud vs on github because we also had to factor in the fact that the ai would need to have access to it. Right now, im woring on user authentication in order to add a layer of security to the project while oke works on some other parts. Our hope is that training the ml wont be too costly moving forward and that we are able to transform the data we get into a format the model can parse comfortably.

Repo Check in of Implementation completed: A paragraph describing what implementation/work has been checked in the repo since the last progress report. Note that work is expected to be periodically checked in into the repo (shared with the instructor) by each team member. Each person must check in their OWN work, and do it periodically each week.

I haven't checked work into the repo because the part I undertook with oke being sourcing the logs and results gotten from the ai came out better on his end than they did on mine and we bothagreed itd be better for the project overall but I have indeed logged hours working on the project. The next part which will be logged will be the authentication and moving forward would be the seamless integration of all parts into one assuming we don't face any other roadblocks.

AI Use Section:

- ***Table of AI Tools and Specific Use:*** *It must contain a table showing tool name (with version number, and account type i.e., free vs. premium), and the specific use of the AI tool (e.g., Loveart AI, starter account free trial, Used for creating app LOGO).* ***Value addition column should be filled in your own words, so you accurately capture the work you have done over and above what was already done by AI***

AI Tool Name	**Version, Account Type**	**Specific feature for which the AI tool was used**	**Value Addition** ***What value did you add over and above what AI did***

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			for you? In your own words
Chatgpt	Free	It was used to create parts of the github repo and helped in creating tables and explanations which helped us better understand the roadmap of the work we would be doing. It helped guide in selecting wether to go with the pcap or csv files and wether we would be okay with cves which we later saw to be not the case	I was able to bring more value to the project by asking questions and trying to search up alternate ways to get the job done since initially we were going to use cves which were compilations of vulnerabilities before we switched to pure logs since they would show day in and day out use of a system before and after an attack

- ***Appendix:*** *Prompt history must all be saved in the appendix*

  Prompt history :

  Session 1: Folder structure (refactor/folder-structure)

  Prompt 1 (too vague)

  how should i organize my project folders

  ChatGPT asked what language and framework, then gave a generic src/, tests/, docs/ layout. Not useful for a course with required deliverables.

  Prompt 2 (adds context)

  its a capstone project, ai threat detection platform. python fastapi backend with a log parser and ml engine, runs in docker. right now everything is in the root folder (app, data, Dockerfile, docker-compose.yml, requirements.txt). my course wants a documents folder and an implementation folder

  ChatGPT proposed DocumentsAndReports/ and Implementation/ with a single backend/ inside. It did not account for a frontend or ML service coming later.

  Prompt 3 (corrects the plan)

  i will also add a web dashboard and separate ml training later so one backend folder wont work. can it be like Implementation/apps/ with each service in its own folder?

  ChatGPT gave Implementation/apps/<service>/ and suggested initial-app for the existing backend. Good fit.

  Prompt 4 (asks for exact steps)

  ok which files go inside initial-app and which stay in the root? give me the git commands, i dont want to break my history

  ChatGPT listed app/, data/, Dockerfile, docker-compose.yml and requirements.txt as moving, with README.md and .gitignore staying at the root. It gave git mv commands.

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Prompt 5 (something went wrong)
i ran the git mv commands and it says "fatal: destination directory does not exist"
ChatGPT explained that git mv does not create parent folders and gave mkdir -p Implementation/apps/initial-app first.
Prompt 6 (README)
write a short readme for the repo. title is AI-Powered Threat Detection Platform. it streams live traffic, detects attacks and anomalies in real time, gives severity scores and suggests actions on a central dashboard
ChatGPT produced a title, description and short feature list. The final README kept only the title and description.

Session 2: ML training (feature/ml-training)
Prompt 9 (first attempt, wrong model family)
add machine learning to ml_engine.py to detect attacks in network traffic
ChatGPT wrote a supervised RandomForestClassifier that needed a labelled Label column and discussed class imbalance. Not what the project proposal described.
Prompt 10 (redirects to the right approach)
my proposal says unsupervised anomaly detection because zero-day attacks wont have labels. use isolation forest instead. dataset is CIC-IDS-2017
ChatGPT rewrote the script with IsolationForest and a contamination parameter.
Prompt 11 (data errors)
getting KeyError: 'Flow Duration' when loading the csv
ChatGPT explained that CIC-IDS-2017 column names have leading spaces and suggested df.columns = df.columns.str.strip().
Prompt 12 (more data errors)
now it crashes with ValueError: Input contains infinity or a value too large for dtype float64
ChatGPT said the dataset contains Infinity and NaN in rate columns, and added replace([np.inf, -np.inf], np.nan) followed by dropna().
Prompt 13 (asks for persistence)
training works. save the trained model to app/models/isolation_forest.joblib so i dont have to retrain every time
ChatGPT added joblib.dump and a load_model() helper.
Prompt 14 (detection logic)
now write a function that loads the saved model and takes one traffic record and returns whether its an anomaly plus a score between 0 and 1
ChatGPT returned a first version that used raw decision_function output. After a follow-up it normalised the score.
Prompt 15 (documentation)
write a readme section for the layer 3 detection engine: what it does, the isolation forest and dataset, the folder structure, and how to run training
Session 3: Dashboard (feature/dashboard-setup)
Prompt 16 (wrong stack)
make me a dashboard for my threat detection platform

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ChatGPT produced a single HTML file with inline CSS and Chart.js. It worked, but it could not grow into a multi-page app.

Prompt 17 (names the stack)

i want this as a proper next.js app with typescript, tailwind and shadcn/ui, put in Implementation/apps/web-frontend. give me the setup commands

ChatGPT gave npx create-next-app@latest options and npx shadcn@latest init, then listed the shadcn components to add (card, badge, button, input, switch, table).

Prompt 18 (first layout attempt)

build the home page. sidebar on the left, top row of kpi cards, a couple of charts, a live log table, and a banner suggesting what action to take. use dummy data

ChatGPT put everything into one very long page.tsx. It rendered, but it was hard to read.

Prompt 19 (asks for structure)

this is too long for one file. split into components: sidebar, kpi-cards, dashboard-charts, log-feed, action-banner. keep the dummy data inside each component for now

Prompt 20 (visual fix)

the log feed table overflows on small screens and the severity badges all look the same. make critical red, high orange, medium yellow, low green and make the table scroll horizontally

Prompt 21 (routing)

the landing page at / should link to /dashboard. add app/dashboard/page.tsx that uses the sidebar layout

Prompt 22 (wrap-up)

write a conventional commit message for adding the dashboard home page, and add how to run the frontend (npm install, npm run dev) to the readme

That produced feat(dashboard): add home page with dummy data and the final README update.

The Grading Rubric for the Progress Report

Criteria	Grading
Work Logs • The consistency of the work logs • The details and description of the work logs • The granularity of the work logs	2 points
Summary Description of Work • The complexity and technical merit of the work done • The documentation of work done and issues logged in the repo	2 points
AI Use Section • AI Use section is specific and detailed • Prompt is included in the appendix	2 points

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Repo Check In • Frequency of work completed • Frequency of repo check in • Quality of work completed	4 points